

ANSWER KEY

Assignment Testing • CSC354

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1 In the context of machine learning, which algorithm is best suited for environments with complex dyn...

' **Answer: B. Option B: Deep Reinforcement Learning**

Explanation: Deep reinforcement learning is a subset of machine learning that combines deep neural networks with reinforcement learning principles to enable agents to learn optimal strategies through trial and error in complex environments.

2 Which algorithm is commonly used to train agents in environments with complex dynamics for reinforce...

' **Answer: B. Reinforcement Learning using Q-learning**

Explanation: Q-learning is a reinforcement learning algorithm that enables agents to learn optimal actions through trial and error in complex environments.

3 Which of the following algorithms is commonly used in deep learning for supervised tasks?

' **Answer: B. Convolutional Neural Networks (CNN)**

Explanation: CNN is a deep learning algorithm that has proven highly effective in tasks involving image recognition and classification.

4 Which algorithm is most suitable for learning policies in environments with complex dynamics?

' **Answer: C. Deep Q-Network (DQN)**

Explanation: Deep reinforcement learning algorithms like Deep Q-Networks are designed to handle high-dimensional state spaces and learn optimal policies through trial and error.

5 In the context of machine learning, which algorithm is best suited for environments with complex dyn...

' **Answer: B. Option B: Deep Reinforcement Learning**

Explanation: Deep reinforcement learning algorithms are designed to handle complex environments by using neural networks as function approximators for the policy and/or value functions. They learn optimal policies through trial-and-error interactions with the environment.

6 In the context of machine learning, which algorithm is commonly used to solve problems in environmen...

' **Answer: C. Reinforcement Learning**

Explanation: Reinforcement learning is a type of machine learning where an agent learns to make decisions by performing actions in an environment and receiving rewards or penalties. It's particularly suited for complex dynamic environments.

7 Which algorithm is often used for learning policies in complex environments where the agent must lea...

' **Answer: B. Neural Networks with Reinforcement Learning**

Explanation: Reinforcement learning algorithms enable agents to learn optimal policies through trial and error by receiving rewards or penalties for actions in complex environments.

8 Which algorithm is most suitable for solving problems where the environment's dynamics are complex a...

' **Answer: C. Reinforcement Learning**

Explanation: Reinforcement learning algorithms are designed to make decisions and learn optimal policies in environments where the dynamics can change over time.

9 Which algorithm is best suited for environments with complex dynamics and requires the model to learn...

' **Answer: C. Q-Learning (a type of Reinforcement Learning)**

Explanation: Reinforcement learning algorithms like Q-learning are designed to make sequences of decisions by interacting with the environment and receiving rewards or penalties. This approach is particularly useful in complex dynamic environments where an agent must learn optimal policies through trial and error.

10 Which algorithm is commonly used for training agents to perform tasks in complex, dynamic environments...

' **Answer: C. Q-Learning (Reinforcement Learning)**

Explanation: Q-learning is a model-free reinforcement learning algorithm that does not require the problem to be modeled explicitly. It can learn optimal policies through trial and error interactions with complex environments.
